Review Test Submission: TSP 01

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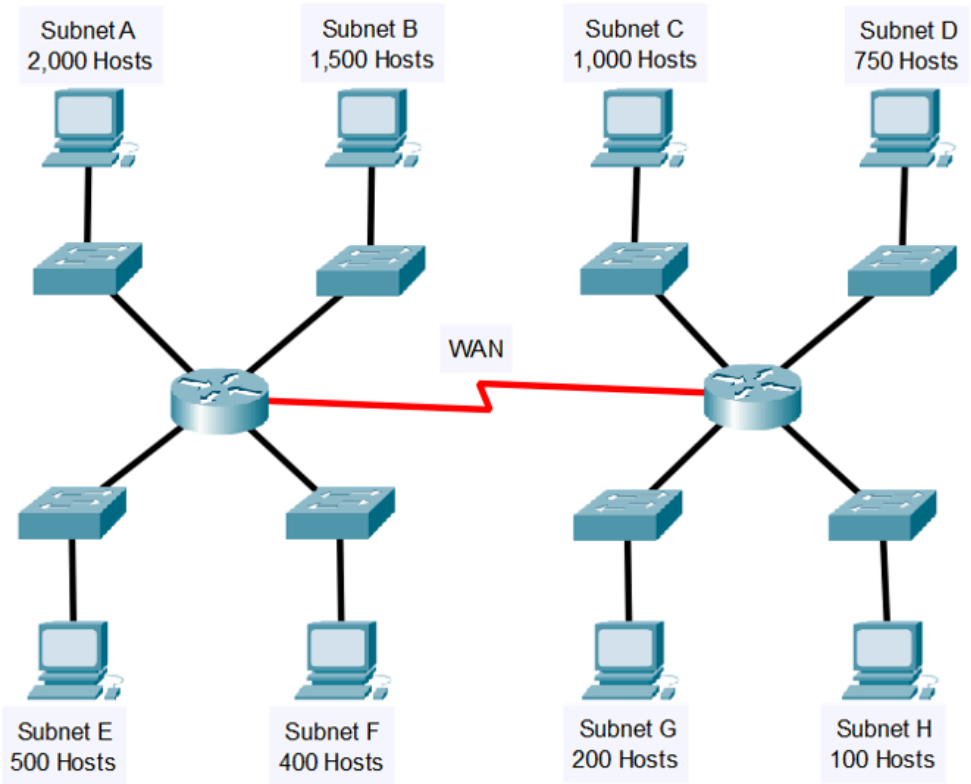
Course	Fall 2024 Switching and Routing (NET-126-2001)
Test	TSP 01
Started	10/31/24 9:24 AM
Submitted	10/31/24 9:54 AM
Status	Completed
Attempt Score	100 out of 100 points
Time Elapsed	29 minutes out of 2 hours
Results Displayed	Submitted Answers, Incorrectly Answered Questions

Question 1

5 out of 5 points



Refer to the exhibit. Use traditional subnetting to develop an addressing scheme based on the number of **subnets** (including the WAN) using 172.16.0.0/16 as the starting point, then provide the requested information below. Include the prefix length for the network addresses only. Express the interval and octet using the format: 32, 4th

Determine the following: **Custom Address Map (CAM)**

Hint: Since the requirements are based on the number of subnets:

1. Use formula for Total Subnets: 2^S
2. Set it greater than or equal to the number of networks shown in the topology inculding the WAN.
3. Solve for s.
4. The value of s represents the number of 's' (subnet/borrowed) bits in the CAM.

Selected Answer: N.N.sssshhhh.H

Question 2 5 out of 5 points

Determine: **Interval and Transition Octet (TO)**

Hint 1: The Interval is the value of the last (rightmost) subnet bit within the octet.

Possible values are: 128, 64, 32, 16, 8, 4, 2, and 1

Hint 2: The Transition Octet is the octet where the last network/subnet bit resides.

Specify the answer using the format: *interval, octet*. For example: 8, 2nd

Selected Answer: 16, 3rd

Question 3 4 out of 4 points

Determine the following: **Prefix Length**

Specify it as /NN, for example: /14

Selected Answer: /20

Question 4 4 out of 4 points

Find: **Custom Subnet Mask**

Hint 1: Refer to the CAM. Turn on the network bits (set to 1) and turn off the host bits (set to 0).

Hint 2: Remember that subnet bits are network bits.

Selected Answer: 255.255.240.0

Question 5 5 out of 5 points

Determine: **Wildcard Mask**

Hint 1: Refer to the CAM. Turn off the network bits (set to 0) and turn on the host bits (set to 1).

Hint 2: Remember that subnet bits are network bits.

Selected Answer: 0.0.15.255

Question 6 4 out of 4 points

Determine: **Total Number of Subnets Possible**

Hint: Total Subnets = 2^s, where s is the number of subnet bits in the CAM.

Selected Answer: 16

Question 7 4 out of 4 points

Determine: **Total Number of Addresses (per subnet)**

Hint: Total Addresses = 2^h, where h is the number of host bits in the CAM.

Selected Answer: 4096

Question 8 5 out of 5 points

Determine: **Total Number of Usable Addresses (per subnet)**

Hint: Total Usable = 2^h - 2, where h is the number of host bits in the CAM.

Selected Answer: 4094

Question 9 4 out of 4 points

Subnet A: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.0.0/20

Question 10 4 out of 4 points

Subnet A: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.15.255

Question 11 4 out of 4 points

Subnet B: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.16.0/20

Question 12 4 out of 4 points

Subnet B: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.31.255

Question 13 4 out of 4 points

Subnet C: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.32.0/20

Question 14 4 out of 4 points

Subnet C: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.47.255

Question 15 4 out of 4 points

Subnet D: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.48.0/20

Question 16

4 out of 4 points

Subnet A: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.63.255

Question 17

4 out of 4 points

Subnet E: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.64.0/20

Question 18

4 out of 4 points

Subnet E: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.79.255

Question 19

4 out of 4 points

Subnet F: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.80.0/20

Question 20 4 out of 4 points

Subnet F: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.95.255

Question 21 4 out of 4 points

Subnet G: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.96.0/20

Question 22 4 out of 4 points

Subnet G: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.111.255

Question 23

4 out of 4 points

Subnet H: Network Address and Prefix Length

Hint 1: Start with the given address to subnet and apply the proper prefix.

Hint 2: Determine all network addresses first.

Record your answer using the following format as an example: 192.16.1.128/27

Selected Answer: 172.16.112.0/20

Question 24

4 out of 4 points

Subnet H: Broadcast Address

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix length.

Selected Answer: 172.16.127.255

Thursday, November 7, 2024 11:26:11 AM EST

← **OK**